

management. The market comprises large players like Rockwell, Schneider, Siemens and several fragmented players who have developed capabilities in specific fields. Rishabh Instruments, over three decades, has developed expertise in electrical equipment and systems like energy management."

Rajeev Sharma, general manager-Strategic Planning and Business Development, Mitsubishi Electric India Pvt Ltd, says, "Automation Industry is increasing at the rate of 12 to 13 per cent every year, but in the Indian automation market technology used in machinery is not at par with developed countries. Indian market is good for F&B and Pharma industry which is a very attractive market for automation."

The job scenario

There is considerable demand for control and automation engineers in India, say industry experts. "The scenario for jobs in control and automation clearly shows a serious demand for students and engineers from the branches of electrical, electronics and instrumentation engineering," says Nilesh Sawant. He adds, "All types of industries, including automobiles, food and beverages, pulp and paper, cement, mining, chemical, pharma, construction, infrastructure or even power generation and transmission, use automation engineering and technology in small, medium or high levels, depending on the machinery or process. Hence, individuals gaining knowledge in basic or advanced levels of automation will have access to a variety of job opportunities."

Considering India's industrial growth, many multinational companies that provide automation solutions have entered or are entering into the Indian market, informs Parmita Kapat. She says, "These companies have to rely on local resources. The manpower can be for office staff (designing and manufacturing) or for onsite deployment. If a skilled workforce is not available locally, these MNCs have to 'import' talent, which increases the cost of the solution." She adds, "There is a large requirement for products

The gist of the training at Siemens

SITRAIN is a training program designed for customers and students in the field of automation and drives. Started in 1980, SITRAIN is aimed at elevating people's skills to adapt to newer challenges. The training module encompasses a wide range of courses designed for CNCs, PLCs, drives, controls, HMI, networks, process control, analysers and instrumentation, electrical and power systems, safety and more. Through specific courses designed for working professionals and students, each course schedule consists of well balanced theory and provides hands-on training to optimise the knowledge about the products.

Siemens conducts the following training programmes:

- a) Basic PLC – 5 days
- b) Basic SCADA – 5 days
- c) Basic Networking – 4 days
- d) Basic DCS – 6 days

Siemens also conducts training in areas related to industry automation like:

- a) Basic Process Instrumentation
- b) Basic AC/DC Drives (Power Electronics & Motor Control)
- c) Basic LV – Switchgear
- d) Basic Mechatronics (Siemens, Berlin-based)

—Courtesy: Nilesh Sawant, head, SITRAIN India, Siemens Ltd

used in control and automation. Most of the raw material used for automation solutions is imported. The industry needs substitutes for the imports."

Kushawaha thinks that those aspiring to enter this field should take up internship opportunities during college, at companies involved in such operations. Plant start-ups are a great place to experience systems integration and functioning. Another avenue for learning are the various competitions held at educational institutes across the country. He says, "Aspirants should get out of their comfort zones and take up such challenges. Rishabh Instruments offers internship opportunities to students in the field of energy management and systems integration. Students get to dirty their hands on actual systems, which leaves them with an intense learning experience."

Scope and entry level roles in the control and automation sector

Automation is a rapidly expanding field and in a developing country like India, there are immense growth opportunities for an individual. Kushawaha says, "An aspect of crucial importance here is that an exposure to operations gives you an edge over others, as it helps you to acquire a holistic picture of the systems involved." Talking about entry level jobs, he says,

"The entry level roles do involve getting into installation, commissioning, etc, and these experiences help one come up with efficient and safe designs in automation projects."

Regarding the scope for career development in this field, Prasad Bhatt informs, "Broadly, an engineer can work in three different kinds of companies. The first are pure-play automation companies that are suppliers of automation equipment and software to various industries. Then there are industrial users of automation products and services, such as companies in automotive, power and the oil & gas sectors. Finally, there are services companies, which provide consultancy and services to integrate products from pure-play automation companies with the manufacturing processes of the industrial users." These services add high value by adapting standalone machines and middleware, and customise them to the specific needs of optimisation and control, remote management and machine-to-machine (M2M) applications. Talking about entry level jobs, he adds, "Typically, entry level roles involve validation of automation and control systems' designs. Gradually, the engineer moves into designing and modelling of the systems."

According to Nilesh Sawant, there are two broad categories of firms that absorb freshers or engineers with one